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Merhdad Saif
Editor-in-Chief
IEEE Access

Dear Editor-in-Chief,

I am writing to submit our manuscript entitled **“Performance Indicators for Shape and Position Assessment in Electromagnetic Inverse Scattering”** for consideration for publication in IEEE Access.

This manuscript presents a contribution to the field of electromagnetic inverse scattering by introducing two novel performance indicators specifically designed to evaluate algorithm performance in terms of shape reconstruction accuracy and object localization precision. Unlike traditional metrics such as Mean Square Error (MSE) or Structural Similarity Index (SSIM), which primarily assess contrast estimation accuracy, our proposed indicators provide direct insight into the recovery of object geometry and spatial position—critical aspects for practical applications.

Key features of this work include:

- Introduction of two novel performance indicators: a shape error indicator and a position error indicator, specifically tailored for electromagnetic inverse scattering evaluation;
- Comprehensive experimental validation using six traditional algorithms: Linear Sampling Method (LSM), Orthogonality Sampling Method (OSM), Born Iterative Method (BIM), Contrast Source Iterative Method (CSI), Subspace Optimization Method (SOM), and Circle Approximation (CA);
- Three distinct experimental frameworks: shape recovery studies, position detection studies, and realistic breast phantom experiments;
- Robust statistical analysis revealing that BIM achieves optimal performance for geometric reconstruction in low nonlinearity scenarios, while SOM and OSM excel in target localization for small, high-contrast scatterers;
- Open-source implementation through the EISPY2D library, ensuring reproducibility and facilitating future research.

The proposed indicators address a critical gap in the current evaluation methodology for electromagnetic inverse scattering algorithms. They are applicable to both qualitative and quantitative methods and can be effectively used when these approaches are combined in experimental studies. This work provides the electromagnetic inverse scattering community with essential tools for algorithm development, benchmarking studies, and performance assessment.

Relevance to IEEE Access:

This work directly aligns with IEEE Access's multidisciplinary scope, addressing fundamental aspects of electromagnetic inverse scattering, computational electromagnetics, and performance evaluation methodologies. The proposed indicators will be valuable for researchers across multiple IEEE fields of interest, including signal processing, biomedical imaging, microwave systems, and computational intelligence. The open-access nature of IEEE Access will ensure broad dissemination of these tools to the international electromagnetic inverse scattering community and facilitate their adoption in diverse application domains.

This manuscript is not under consideration by any other journal. The authors declare no conflicts of interest and confirm that this work contains original research. All experiments were conducted ethically, and proper attributions have been made to previous work.

Thank you for your time and consideration. We look forward to your response and are available to address any questions or concerns regarding our submission.

Sincerely,

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